L4 – Transformations of Sine and Cosine Part 2

MCR3U

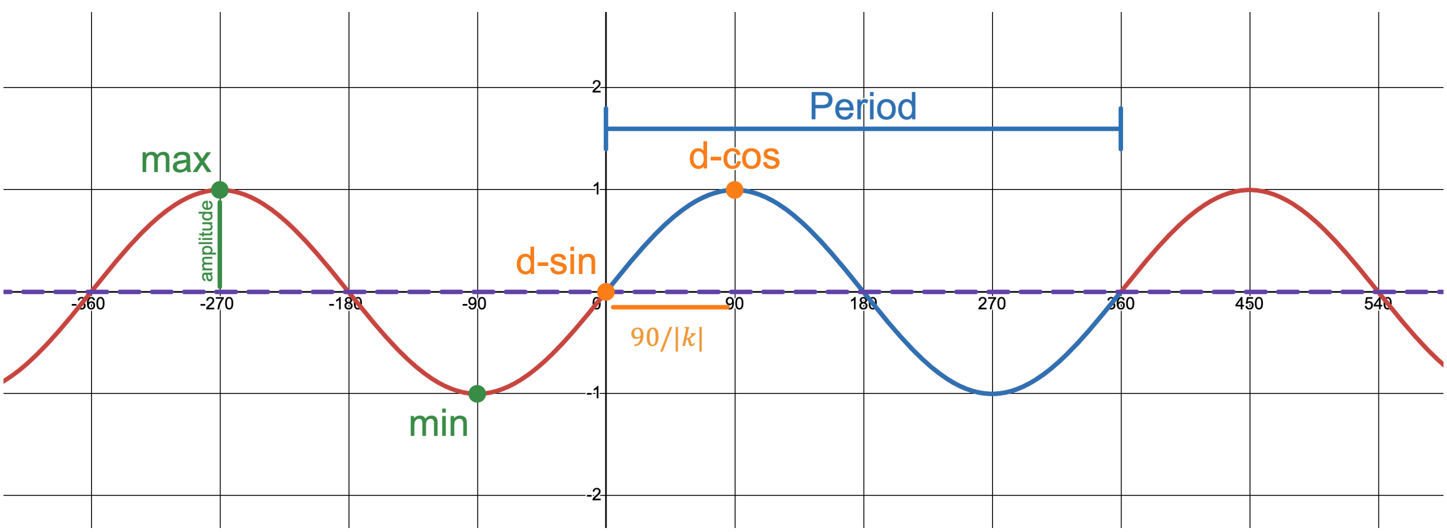
Graph → Equation

How to Determine the Equation of a Sine or Cosine Function Given its Graph

To determine the equation of a transformed sine or cosine function, you must use the graph to determine the value of the parameters $a, k, d,$ and c to then write the transformed equation in the format:

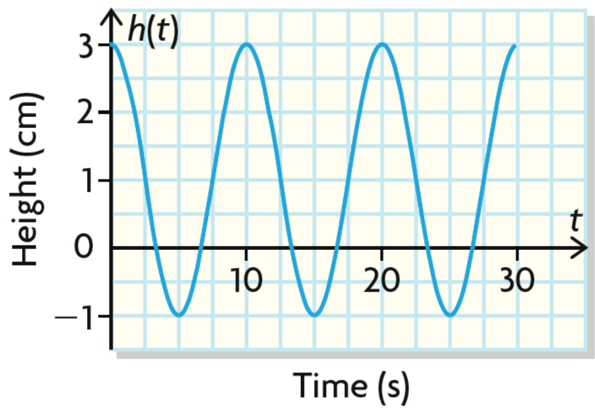
$$y = a\sin[k(x-d)] + c \qquad \text{OR} \qquad y = a\cos[k(x-d)] + c$$

a	k	d	c
Find the amplitude of the function:	Find the period (in radians) of the function using a starting point and ending point of a full cycle.	**for $\sin x$:** x-coordinate of a rising mid-line. **for $\cos x$:** x-coordinate of a maximum point.	Find the vertical shift OR (this finds the 'middle' of the function)

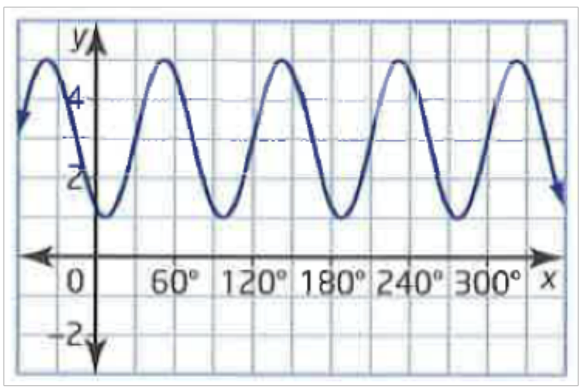


Example 1: For each of the following graphs, determine the equation of a sine and cosine function that represents each graph:

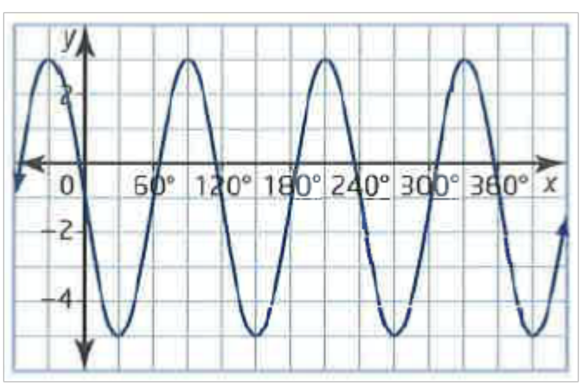
a)



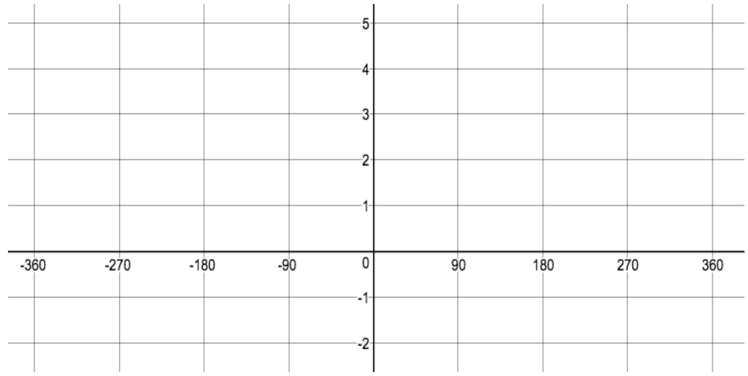
b)



c)



Example 2: A sinusoidal function has an amplitude of 3 units, a period of 180 degrees and a max point at (0, 5). Represent the function with an equation in two different ways.



Example 3: A sinusoidal function has an amplitude of 5 units, a period of 120 degrees and a maximum at (0, 3). Represent the function with an equation in two different ways.